

Module 1 summary cheat sheet

Function definitions:

```
def FuncName (arg):  
    some_code()  
    # comment  
    return var_name  
end
```

Arg and return are optional

If Statements:

```
if (var==True):  
    some_code()  
elif (a==b):  
    some_code()  
else:  
    some_code()  
end
```

For loops:

```
i=0  
k=5  
while (i<k):  
    some_code()  
    i=i+1  
end
```

Module 2 summary cheat sheet

Functions:

- `popup(string_var, title = "title string", blocking=True)`
- `to_str()` also `to_num()`
- `request_float_from_primary_client("text to show")` and `string/boolean/integer` instead of float
- `sleep(t)`
- `sync()`

Module 3 summary cheat sheet

Functions to work with lists:

- `Var.length()`
- `Var.capacity()`
- `Var.excess_capacity()`
- `Var.to_string()`
- `Var.slice(index1, index2)`
- `Var.append(element)`
- `Var.extend(list of elements)`
- `Var.insert(index, element)`
- `Var.pop()`
- `Var.remove(index)`
- `Var.clear()`

Functions to create lists and structs:

- `Var=make_list(length, initial_value, capacity)`
- `Var=struct(identifier1=value1, id2=value2, id3=v3)`

Functions to use with matrices:

- `Var.get_column(index)`
- `Var.get_row(index)`
- `Var.shape()`

Module 4 summary cheat sheet

Functions:

- `move(pose, a, v, r)` and `optimovel(pose, a, v, r)`
- `movej(pose, a, v, r)` and `optimovej(pose, a, v, r)`
- `movep(pose, a, v, r)`

- `get_actual_tcp_pose()`
- `pose_add(p1, p2)`
- `pose_sub(p1, p2)`
- `pose_trans(p1, p2)`
- `pose_inv(pose)`
- `pose_dist(p1, p2)`
- `point_dist(p1, p2)`

Module 5 summary cheat sheet

Define

```
thread thread_name():  
    while True:  
        # code to loop  
        sync()  
    end  
end
```

Run

```
control_var=run thread_name()
```

Kill

```
kill control_var
```

Wait

```
join control_var
```

Critical

```
enter_critical  
exit_critical
```

Module 6 summary cheat sheet

Functions:

- `freedrive_mode(axes, frame)`
- `end_freedrive_mode()`

- `zero_ftsensor()`
- `is_steady()`

- `force_mode(task_frame, selection_vector, wrench, type, limits)`
- `end_force_mode()`

- `force()`
- `get_tcp_force()`

Module 7 summary cheat sheet

Functions:

- `get_forward_kin(joints)`
- `get_inverse_kin(pose, qnear)`
- `get_inverse_kin_has_solution(pose)`
- `is_within_safety_limits(pose)`
- `interpolate_pose(p_from, p_to, alpha)`

Module 8 summary cheat sheet

Functions:

- `speedj(qd, a, t)`
- `stopj(a)`
- `speedl(xd, a, t)`
- `stopl(a)`
- `servoj(q, t, lookahead_time, gain)`

Module 9 summary cheat sheet

Functions:

- `add_frame(name, pose, ref_name="base")`
- `attach_frame(child, parent)`
- `delete_all_frames()`
- `delete_frame(name)`
- `move_frame(name, pose, ref_name)`
- `get_pose(name, rel_name="base", ref_name=rel_name)`
- `convert_pose(pose, from_frame, to_frame)`
- `reset_world_model()`
- `is_frame(name)`
- `get_frame_parent(name)`
- `get_children(name)`